Kamiar Asgari

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Research Profile

Ph.D. in Electrical Engineering from the University of Southern California (October 2026). Research at the intersection of machine learning and optimization, focusing on sequential decision-making and large-scale, real-time optimization in adversarial and stochastic environments. Strong theoretical background in machine learning, linear algebra, and statistics.

Work authorization: Currently on F-1 OPT. Open to relocation worldwide.

Education

Aug 2018 – Oct 2026 **University of Southern California (USC)**
Los Angeles, CA
Ph.D., Electrical Engineering
Minor: Industrial and Systems Engineering
GPA: 3.90/4.00

USC coursework

- **Optimization, probability & networks:** Large Scale Optimization and Machine Learning (ISE 633); Stochastic Network Optimization (EE 649); Data Networks: Design and Analysis (EE 550); Foundations of Stochastic Processes (ISE 620); Random Processes in Engineering (EE 562); Probability (EE 503).
- **Machine learning:** Machine Learning I: Supervised Methods (EE 559); Machine Learning II: Mathematical Foundations and Methods (EE 660); Machine Learning for Data Science (DSCI 552); A Computational Introduction to Deep Learning (EE 541).
- **Algorithms, mathematics & signals:** Analysis of Algorithms (CSCI 570); Fundamental Concepts of Analysis (MATH 425A); Quantum Error Correction (EE 514).

Aug 2018 – May 2024 **University of Southern California (USC)**
Los Angeles, CA
M.S., Electrical Engineering

Sep 2014 – Jul 2018 **Sharif University of Technology**
Tehran, Iran
B.S., Electrical Engineering (Communications)
GPA: 17.99/20

Selected coursework: Computer Structure and Microprocessors & Lab, Signals & Systems, Digital Signal Processing, Digital Communications, Linear Control Systems, Engineering Probability and Statistics, Numerical Computation.

Industrial Training: Summer of the 2016–2017 academic year; rated Excellent.

Honors

2023 **USC–Amazon Best Poster Award**
For nonsmooth projection-free optimization research at the USC/Amazon Machine Learning Center's second symposium.

2014 **National Elite Foundation of Iran Fellowship**
Full undergraduate scholarship.

2013 **Iranian National Physics Olympiad – Bronze Medalist**
23rd edition; top 50 nationwide; theoretical physics and problem-solving. Awarded eight physics and laboratory credits at Sharif University of Technology for the Olympiad achievement.

National University Entrance Exam (Konkur)
Ranked 105th out of 150,000+ students (top 0.1%).

Industry Experience

Oct 19, 2026 – Present **Yumi Health**
Software Engineer

May – Jun 2026 **Diligence Machines**
Redwood City, CA
PhD AI Fellow

- **LLM optimization:** Balanced workloads across multiple LLM calls to prevent bottlenecks. Moved mathematical calculations from the LLM to the backend, fixing calculation errors and reducing API costs.
- **System reliability:** Fixed a critical bug that was losing 30% of system emails by implementing robust retry logic. Discovered and diagnosed system failures through automated local tests in Docker.
- **Text clustering:** Proposed upgrading a regex-based text parser to LLM-driven atomic proposition extraction to improve topic grouping.
- **Applied AI:** Worked on LLM prompt optimization to streamline document analysis for M&A due diligence. Developed an LLM-based collaboration tool to automate the aggregation and summarization of team brainstorming sessions and decision workflows.

May – Aug 2025 **Chevron CiSoft (USC)**
Los Angeles, CA
Research Intern – Optimization & ML
Supervisor: Dr. Andrei Sergiu Popa

- **Forecasting and predictive modeling:** Trained neural networks on historical production data to forecast two-year cumulative barrels of oil equivalent (BOE), achieving test $R^2 = 0.782$. Managed data curation and feature engineering to support operational planning and resource allocation.
- **Neural surrogates and inverse design:** Trained PyTorch neural networks to approximate complex physical simulations, with test $R^2 = 0.865$ for the inverse-design surrogate. Built

multi-output MLPs with scaling and early stopping. Designed multi-objective hydraulic-fracturing inverse-design pipelines, combining discretized inverse mapping with quasi-Newton refinement to recover diverse designs. Used neural surrogates as objective functions for evolutionary methods, including genetic algorithms, as well as particle swarm optimization and gradient descent, balancing cost, production, and operational constraints. Code: neural-frack-optimizer.

- **Spatial optimization:** Developed a two-phase pipeline combining exponential and binary search with derivative-free optimizers to maximize the number of infills under strict geometric constraints.
- **Geometric optimization:** Developed a Voronoi-based spatial optimization algorithm in Python using Shapely and NumPy to automate facility placement and generate spatial topologies under boundary, spacing, density, and coverage constraints. Used ε-grid and iterative Voronoi-diagram refinement; evaluated Differential Evolution, PSO/QPSO, and genetic algorithms for constrained layout search. Code: Voronoi-for-oil.
- **Visualization and decision support:** Built an interactive 3D response-surface explorer in Plotly for proppant/fluid design, spatial analysis, scenario modeling, and stakeholder trade-off analysis. Designed metrics, conducted exploratory data analysis and sensitivity studies, and iteratively updated suggested designs using domain-expert feedback to inform planning recommendations.
- **Experimental evaluation:** Benchmarked optimizers through controlled experiments. Analyzed end-to-end decision-pipeline latency and throughput, using the results to adjust batching and ordering. Improved surrogate throughput by up to 15× through memory-efficient batching and optimized I/O.
- **Research software:** Delivered modular, tested, documented Python packages with data loaders, training loops, scalers, optimization routines, configuration, logging, and reproducible runs. Automated Python/SQL data preparation with checks for missing values, ranges, and duplicates. Collaborated with engineers and researchers to turn operational questions into modeling and optimization tasks.

The inverse-design and geometric-optimization work was presented at the 2026 SPE Western Regional Meeting.

Additional project code: Eagle-Ford.

Research Experience

Aug 2019 – Oct 2026 **University of Southern California**
Los Angeles, CA
Graduate Researcher
Advisor: Prof. Michael J. Neely

Additional collaboration with Dr. Stephen Tu included continual learning.

- **Real-time allocation and control:** Engineered online policy selection over linear FIR filters to maximize utility under worst-case arrivals and adversarial, state-dependent utility functions. Used time-varying moving-average filters to map arrival time series to resource allocations. Developed optimization algorithms with mathematically proven logarithmic scalability to minimize costs under adversarial resource arrivals. Extended the POMACS 2020 framework,

presented at SIGMETRICS 2021, to non-stochastic inputs and costs dependent on both storage state and allocation. Applications include energy storage, queuing networks, and inventory management. Code: adversarial online control.

- **Finite-capacity storage policies:** Developed simplex-constrained disturbance-action policies (SDAC), enforcing nonnegative allocations, inventory conservation, and physical feasibility in a finite buffer. Established $O(\sqrt{T})$ regret against the best fixed SDAC policy under convex Lipschitz costs and bounded disturbances. Built Python simulations to validate theoretical bounds and benchmark algorithms against standard solvers.
- **Energy-harvesting online optimization:** Developed Bregman-divergence online learning methods with sublinear-regret guarantees against adaptive adversarial losses while respecting physical energy availability. The model uses i.i.d. resource arrivals and convex, allocation-only losses, comparing performance with the best fixed allocation in hindsight. Used two distinct Bregman divergences and generalized multiplicative updates to storage systems with random resource arrivals and memory, connecting online learning with max-weight and queueing ideas. This first-author work was published in *POMACS* (2020) and presented at *ACM SIGMETRICS* (2021).
- **Constrained and nonsmooth optimization:** Resolved an open problem by generalizing Frank–Wolfe to nonsmooth objectives and functional constraints, combining ADMM with primal-dual methods. Developed a projection-free framework for large-scale Lipschitz nonsmooth convex problems with many functional inequality constraints, using stochastic first-order/subgradient and linear-optimization oracles. Attained an optimal $O(1/\sqrt{T})$ convergence rate, equivalently $O(\varepsilon^{-2})$ oracle complexity. Used min–max reformulation, Fenchel duality, and optimistic mirror-descent/online-gradient updates. Reduced overhead and accelerated sparse and robust regression, supported by proofs and empirical validation. First-author work appeared in *Computational Optimization and Applications* (2024).
- **Bandit-based reinforcement learning and rate adaptation:** Designed and implemented UCB-based multi-armed bandit algorithms for real-time rate selection and throughput optimization in queuing networks. Learned uncertain random capacities from partial ACK/NACK feedback; transmissions succeed only when the selected rate does not exceed instantaneous capacity. Used discretized-rate UCB, recursive capacity estimation, and stage-wise learning/control analysis to establish uniform-in-time backlog bounds. Analyzed latency–throughput tradeoffs and delayed-reward settings, with applications to dynamic pricing under inventory constraints.
- **Research methods:** Developed drift and regret proofs, synthetic and stochastic simulations, ablation studies, and benchmark comparisons. Implemented vectorized, numerically stable projection-free optimization in Python, demonstrating a 3× speedup over projected-gradient methods on high-dimensional benchmarks. Communicated findings through lab presentations, manuscripts, and sponsor briefings.

Summer 2024 **STARS Lab, University of Southern California**
Los Angeles, CA
Research Assistant – Theory of Next-Token Prediction
Advisor: Dr. Stephen Tu

- Investigated single-layer Transformers and connections to contextual online logistic regression and Universal Portfolio Theory. Demonstrated the reduction under stochastic gradient descent and analyzed regret using Online Mirror Descent (OMD) and Follow-the-Regularized-Leader (FTRL).
- Related the analysis to sequential asset allocation, logarithmic wealth growth, time-series forecasting, and attention interpretability. Studied aggregation, improper learners, and computational versus regret-optimal tradeoffs; identified dimension-dependent bottlenecks and explored more efficient algorithms.
- Explored self-concordant barriers, interior-point methods, log-concave sampling, and connections to optimal transport and natural gradients. Implemented experimental PyTorch pipelines for online logistic regression and Transformer analysis with streaming data.

Aug 2018 – May 2019 **Cyber-Physical Systems Group, USC**
Los Angeles, CA
Research Assistant
Advisor: Dr. Paul Bogdan

- Analyzed space–time fractional diffusion using data-driven methods; identified parameters and fitted fractional PDEs to trajectory time series to model long-term dependencies in non-classical diffusion.
- Contributed to *Identifying Arguments of Space–Time Fractional Diffusion: A Data-Driven Approach* (2020), with M. R. Znaidi, G. Gupta, and P. Bogdan.

Aug 2017 – Apr 2018 **Optical Networks Research Laboratory**
Sharif University of Technology, Tehran, Iran
Undergraduate Research Assistant
Advisor: Prof. Jawad A. Salehi
Developed and evaluated DNA sequence alignment using optical parallel processing techniques.

Summer 2016 **Advanced Communications Research Institute**
Sharif University of Technology, Tehran, Iran
Undergraduate Research Intern
Advisor: Prof. Farokh Marvasti
Explored matrix completion and compressed sensing using convex optimization methods.

Selected Projects

AI Checklist Compliance & Gap Analysis System

- Developed a two-stage LLM-powered pipeline using the Gemini API (Flash-Lite and Pro) and Pydantic for document relevance filtering and multi-document gap analysis.
- Implemented schema-constrained structured outputs, automated metadata validation (including date and entity checks), and cost-optimized candidate pruning for scalable checklist evaluation.

American Sign Language Image Classification GitHub

USC EE541, Computational Introduction to Deep Learning, Fall 2022. Instructor: Dr. Brandon Franzke; collaborator: Mahdi Sajedi.

- Built an end-to-end CNN pipeline in PyTorch for 87,000 images across 29 classes, including data preprocessing, augmentation, and custom data loaders. Achieved 97.6% test accuracy (97.56% before rounding), with 99.64% training and 97.41% validation accuracy.
- Built a reproducible CUDA training pipeline using SGD with momentum, mixed precision, and StepLR learning-rate scheduling on an NVIDIA Quadro RTX 5000; profiled approximately 5× training throughput relative to CPU execution. Used Torch-Summary alongside PyTorch and the Python data-analysis libraries.
- Evaluated precision, recall, and F1 scores, confusion matrices, per-class accuracy, and error breakdowns to examine model behavior and failure modes.

Bird Species Classification GitHub

USC DSCI552, Machine Learning for Data Science, Summer 2023. Instructor: Dr. Mohammad Reza Rajati.

- Fine-tuned EfficientNetB0 and VGG16 to classify 20 bird species using Keras and OpenCV, transfer learning, data augmentation, batch normalization, dropout, the Adam optimizer, and early stopping.
- Reached 94.9% test accuracy with EfficientNetB0 (98.1% training/validation accuracy). VGG16 achieved 89.6% test and 96.2% training/validation accuracy. Reported precision, recall, and F1 scores to assess performance on unseen data.
- Explored multimodal embedding fusion.

Numerical Solvers and Dynamic Simulation Tools

- Implemented ODE/PDE solvers and linear-algebra routines in C++; profiled and optimized memory use for scalability on high-performance computing clusters.
- Developed parameterized MATLAB/Simulink models for stochastic-process simulation and estimator testing in energy systems.

Fair-Odds and Expected-Value Modeling Prototype

- Implemented logistic and isotonic calibration and implied-probability conversion; evaluated predictions with Brier score and log loss.
- Explored the sensitivity of expected-value estimates to miscalibration and correlation.

Portfolio Construction and Backtesting Prototype

- Simulated portfolio baskets using mean–variance optimization and simple Kelly-style sizing; backtested rebalancing rules and tracked Sharpe ratio, drawdown, and turnover.

Bandit Recommenders and Discrete Design Choices

- Implemented UCB-style ranking and recommenders for discrete choices and design actions, including exploration penalties and position-bias adjustments.
- Evaluated policies offline using inverse propensity scoring and counterfactual replay to study exploration–exploitation tradeoffs.

Modular Experimentation Platform

- Built modular training orchestration with data loaders, scalers, and cross-validation; combined metric evaluation, calibration, confidence intervals, and reproducible reports.

Distributed Spatial Simulation – Experimental Research Software GitHub

- Developed a distributed simulation system for large-scale spatial optimization using Voronoi partitioning and Differential Evolution, particle swarm optimization, and genetic algorithms.

Distributed Bandit Controller for Networked Queues – Prototype

- Developed an experimental adaptive controller using asynchronous message passing and parallel agents to stabilize networked queues, combining online learning with distributed scheduling.

Teaching & Mentoring

University of Southern California
Los Angeles, CA
Graduate Teaching Assistant

Selected course assignments

- **Fall 2024:** Optimization for the Information and Data Sciences (EE 588); instructor: Prof. Michael J. Neely.
- **Spring 2023 and Spring 2024:** Data Networks: Design and Analysis (EE 550); instructor: Prof. Michael J. Neely.
- **Fall 2023:** Probability for Electrical and Computer Engineers (EE 503); instructor: Prof. Konstantinos Psounis.
- Analysis of Algorithms (CSCI 570), Queueing Networks, and Optimization for Machine Learning.

Head Teaching Assistant — Probability for Electrical and Computer Engineers (EE 503): Managed a team of five TAs and graders.

Graduate mentoring: Mentored M.S. students in Optimization and Algorithms, translating theoretical concepts into implementable code.

Viterbi Graduate Mentorship Program
University of Southern California
Three-time participant providing guidance to graduate students, including two incoming Electrical Engineering master's students in 2023.

Additional coursework: Improvisation and Theatre Games (THTR 122), completed twice.

Professional & Community Service

2024 – 2025 **Journal & Conference Reviewer**
SIAM Journal on Optimization (SIOPT); *Learning for Dynamics and Control* (L4DC).

2024 **USC RoboSub – Software Group**
Participated in software-group work on an autonomous underwater vehicle for the annual RoboSub competition.

2018 **USC Volunteer Center**
Volunteered in Koreatown neighborhood cleanup projects.

Viterbi Impact Program
Supported STEM educational outreach, including MESA events through the Viterbi K–12 Center and the Hustle N' Code Hackathon.

Publications & Manuscripts

In preparation **Adversarial Online Control for Finite-Capacity Storage**
Kamiar Asgari and Michael J. Neely. Manuscript in preparation.

Under review **Bandit-Based Rate Adaptation for a Single-Server Queue**
Stochastic Systems (under review).

2026 **Automatic Identification of Optimum Infill Well Placement using Voronoi Networks and Metaheuristic Optimizers**
First author. *SPE Western Regional Meeting*.

2026 **Intelligent Systems Approach for Optimum Fracturing Treatment Design**
First author. *SPE Western Regional Meeting*.

2024 **Nonsmooth Projection-Free Optimization with Functional Constraints**
Kamiar Asgari and Michael J. Neely.
Computational Optimization and Applications (COAP), 89(3), 927–975.
Preprint.

2020 **Bregman-Style Online Convex Optimization with Energy-Harvesting Constraints**
Kamiar Asgari and Michael J. Neely.
Proceedings of the ACM on Measurement and Analysis of Computing Systems (POMACS), 4(3), 52:1–52:25. Author PDF.

May 2020 **Identifying Arguments of Space–Time Fractional Diffusion: A Data-Driven Approach**
M. R. Znaidi, G. Gupta, Kamiar Asgari, and Paul Bogdan.
Frontiers in Applied Mathematics and Statistics.

Invited Talks & Presentations

2026 **Invited Speaker, 46th SoCal Control Workshop**
University of California, Irvine
Presented *Real-Time Allocation Control for Storage Devices and Inventory Facilities*.

2023 **Selected Presenter, INFORMS Annual Meeting**
Invited presentation on *Nonsmooth Projection-Free Optimization*.

2021 **ACM SIGMETRICS**
Presentation of *Bregman-Style Online Convex Optimization with Energy-Harvesting Constraints*, published in POMACS in 2020.

Technical Skills

Programming & systems: Python, C (basic), C++, SQL (working knowledge of data extraction and joins), Bash, MATLAB/Simulink, R, Go (Golang), Verilog (basic), LaTeX, Git, Linux/Unix, WSL, Docker, VS Code, Jupyter, Excel, ROS.

Numerical & data libraries: NumPy, SciPy, Pandas, Scikit-Learn, Statsmodels, Matplotlib, Seaborn, Plotly, Shapely/GEOS, CGAL, PyTorch, JAX, Keras, OpenCV, Torch-Summary.

Performance & infrastructure: CUDA, Hugging Face Accelerate, multi-GPU training, mixed precision, GPU memory management, CUDA zero-copy, performance profiling and benchmarking, vectorization, concurrency, multithreading, parallelization, TCP/UDP, query-optimization concepts; TensorRT (familiar). AWS S3/Lambda (basic familiarity), Glue/Athena (familiar).

Machine learning & AI: Predictive modeling (linear, ridge, lasso, logistic and probit regression; classification; SVMs and kernel methods; clustering; PCA; anomaly detection); deep learning (CNNs, MLPs, Transformers); transfer learning; neural surrogates; recommendation systems; knowledge graphs; fuzzy logic; large language models (LLMs); prompt engineering; Gemini API; Pydantic; feature engineering, model selection, and evaluation.

Optimization & operations research: Large-scale convex and non-convex optimization, stochastic, robust and distributionally robust optimization, convex–concave minimax, ADMM, primal-dual methods, Frank–Wolfe, linear/quadratic programming (LP/QP), mixed-integer programming (MIP/MILP), interior-point and quasi-Newton methods, coordinate descent, Lagrangian

duality, dynamic programming, Bayesian optimization, heuristic and genetic algorithms, Differential Evolution, PSO/QPSO, Tabu Search, assignment and matching, vehicle routing, resource allocation; Gurobi/GurobiPy, CPLEX, Pyomo, CVXPY, CVX.

Online learning & reinforcement learning: Online mirror descent (OMD), online gradient descent (OGD), follow the leader (FTL), follow the regularized leader (FTRL), entropy regularization, regret analysis, adversarial learning, multi-armed and contextual bandits (UCB, Thompson sampling, EXP3), Markov decision processes (MDPs), policy/value iteration, dynamic pricing, offline policy evaluation.

Statistics & experimentation: Experimental design, A/B and switchback testing, hypothesis testing, causal inference (difference-in-differences and synthetic controls), Bayesian inference, metric design, uncertainty quantification, confidence intervals, cross-validation, ablations, experiment tracking, analysis of experiment results; ROC-AUC, PR-AUC, Brier score, log loss, isotonic calibration.

Control & spatial methods: Real-time and non-stochastic control, stochastic control, Lyapunov/queueing control, model predictive control (MPC), Kalman filtering, digital signal processing (DSP), signals and systems, system dynamics, geometric optimization, Voronoi networks, KD-trees, path planning.

Quantitative analysis & simulation: Stochastic processes, Markov models, queueing models, delay and congestion analysis, service-system stability, Monte Carlo methods, discrete-event simulation, portfolio optimization, universal portfolio theory, mean–variance optimization, risk-averse resource allocation, energy-harvesting models.

Mathematical foundations: Linear algebra (SVD and spectral analysis), calculus, convex analysis, probability theory, concentration inequalities, martingale theory, game theory, graph theory, data structures and algorithms, stochastic network optimization; study of self-concordant barriers, log-concave sampling, optimal transport, and natural-gradient methods.

Forecasting & analytics: Time series and probabilistic forecasting, scenario modeling, capacity planning, utilization optimization, reporting dashboards, ad hoc analysis, model validation, error analysis, data-quality checks, reproducible pipelines, unit testing.

Languages & Additional Interests

Also known as: Kamy.

Languages: Persian (Farsi), native; English, fluent.

Communication: Technical writing and reporting, research presentations, teaching and mentoring, stakeholder updates, and client/sponsor briefings; explaining complex concepts to varied audiences.

Personal interests: Physics, history, economics (including ancient economies), philosophy, stand-up comedy, improvisational theater, and soccer.

Independent study: Financial and energy markets, derivatives and options, market structure, and ERCOT pricing and congestion.

Research Interests & Proposed Directions

Proposed research directions.

- **Adaptive online optimization:** Meta-OCO and policy-parameterized algorithms that learn step sizes, regularization strengths, momentum, exploration rates, and policy families. Nonlinear OCO with predictions, multiple hints, and periodic structure; learning-augmented matching and resource allocation with capacity and fairness constraints.
- **Robust control:** Regret-optimal analogues of LQG, self-tuning regulators, and Gaussian-process model predictive control; robust decision-making under model misspecification and adversarial disturbances.
- **Energy and vehicle-to-grid coordination:** Proposed decentralized primal-dual market mechanisms, robustness to communication delays and selfish agents, extreme-weather stress tests, and safety filters enforcing voltage and battery-capacity constraints.
- **Collective intelligence:** Proposed probabilistic and dynamical models of biological and social coordination, including collaborative foraging as a distributed bandit problem with safety constraints and environmental shocks.
- **AI alignment and structured learning:** Proposed game-theoretic OCO and partial-feedback bandit approaches to Eliciting Latent Knowledge, honesty incentives, and safe exploration. Interests include reasoning pipelines, text-to-database learning, human-feedback meta-optimization, and sequential decision systems.
- **Prediction-market modeling:** Proposed sentiment-gap models linking public communications to Kalshi/Polymarket prices, with aligned text/market data, historical backtesting, and potential execution optimization.
- **Optimization and learning theory:** Bayesian and multi-objective optimization, fairness and imbalanced datasets, nonsmooth nonconvex optimization, efficient LLM training/inference, and state-space models. Broader interests connect deep learning with nonequilibrium thermodynamics and approximation theory.

References

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